Business Statistics

Module 1
Video 1.1
Origin

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Agenda

- Origin of ‘Statistics’

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Agenda

- Origin of ‘Statistics’
- Early forms of data collection

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Agenda

- Origin of ‘Statistics’
- Early forms of data collection
- Early forms of statistical inference

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Agenda

- Origin of ‘Statistics’
- Early forms of data collection
- Early forms of statistical inference
- Timeline of development of statistical methods

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Agenda

- Origin of ‘Statistics’
- Early forms of data collection
- Early forms of statistical inference
- Timeline of development of statistical methods
- Statistics today

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Origin of the word ‘Statistics’

- Thought to have been derived from the Latin word *‘statisticum collegium’* meaning *council of state*

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Origin of the word ‘Statistics’

- Thought to have been derived from the Latin word *‘statisticum collegium’* meaning *council of state*
- The definition was limited to the collection of economic and demographic data
- In the 19th century the definition was broadened and included collection, summarization, and analysis of data

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Early forms of data collection

- First forms of data collection- Crosses, ticks marks and tallies

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Early forms of data collection

- First forms of data collection- Crosses, ticks marks and tallies
- Ancient Egypt- The Nilometer

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  - Measuring the flood levels in the Nile river

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- Roman Empire and Han Dynasty

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Early forms of data collection

- First forms of data collection- Crosses, ticks marks and tallies
- Ancient Egypt- The Nilometer
  - Measuring the flood levels in the Nile river
  - Predicting harvest and taxes
- Roman Empire and Han Dynasty
  - Collected data on the size of the their population, geography, and wealth

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Early forms of statistical inference

- Estimating the required height of ladder- (Athenians)

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Early forms of statistical inference

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  - Several soldiers counted the number of bricks

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Early forms of statistical inference

- Estimating the required height of ladder- (Athenians)
  - Several soldiers counted the number of bricks
  - Multiplied the height of the bricks with the most frequent count

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- Demographic studies

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Early forms of statistical inference

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- Demographic studies
  - John Graunt and William Petty estimated the population of London

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- Estimating the required height of ladder- (Athenians)
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- Demographic studies
  - John Graunt and William Petty estimated the population of London
  - Used measures such as death rates, number of funerals, average family size

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Timeline of development of statiscial methods

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Timeline of development of statiscial methods

17th Century

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Timeline of development of statiscial methods

17th Century

1654

Mathematical Theory of Probability

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Timeline of development of statiscial methods

17th Century

1654

Mathematical Theory of Probability

1662

John Graunt's inferential study on death rates in London

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Timeline of development of statiscial methods

17th Century

1654

Mathematical Theory of Probability

1662

John Graunt's inferential study on death rates in London

1693

Mortality Table- Edmond Halley statistically relates death rate to age

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Timeline of development of statiscial methods

18th Century

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Timeline of development of statiscial methods

18th Century

1713

First derivation of the Law of Large Numbers

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Timeline of development of statiscial methods

18th Century

1713

First derivation of the Law of Large Numbers

1773

Normal Distribution is introduced

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Timeline of development of statiscial methods

18th Century

1713

First derivation of the Law of Large Numbers

1773

Normal Distribution is introduced

1761

Thomas Bayes proves Bayes' Theorem

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Timeline of development of statiscial methods

19th Century

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Timeline of development of statiscial methods

19th Century

1805
Method of least squares- used for fitting curves on given observations

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Timeline of development of statiscial methods

19th Century

1805
Method of least squares- used for fitting curves on given observations

1880
Thiele introduces the Likelihood function

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Timeline of development of statiscial methods

19th Century

1805
Method of least squares- used for fitting curves on given observations

1880
Thiele introduces the Likelihood function

1880
Francis Galton introduces the concept of correlation

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Timeline of development of statiscial methods

20th Century

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Timeline of development of statiscial methods

20th Century

1908

Student's t-distribution is published

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Timeline of development of statiscial methods

20th Century

1908

Student's t-distribution is published

1933

Kolmogorov publishes axioms of probability

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Timeline of development of statiscial methods

20th Century

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Neyman introduces the concept of confidence intervals

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Kolmogorov publishes axioms of probability

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Neyman introduces the concept of confidence intervals

1979

Bradley Efron publishes literature on the Bootstrap Method

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Statistics today

- Evolving technology and Big Data

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Statistics today

- Evolving technology and Big Data
  - Large amounts of data- both quantity and frequency

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- Development of new techniques

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  - Machine learning, deep learning, AI

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  - Several important business decisions are based on statistical evidence

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  - Several important business decisions are based on statistical evidence
  - Statistics like Unemployment rate, GDP, etc. play important roles in economic policy decisions

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